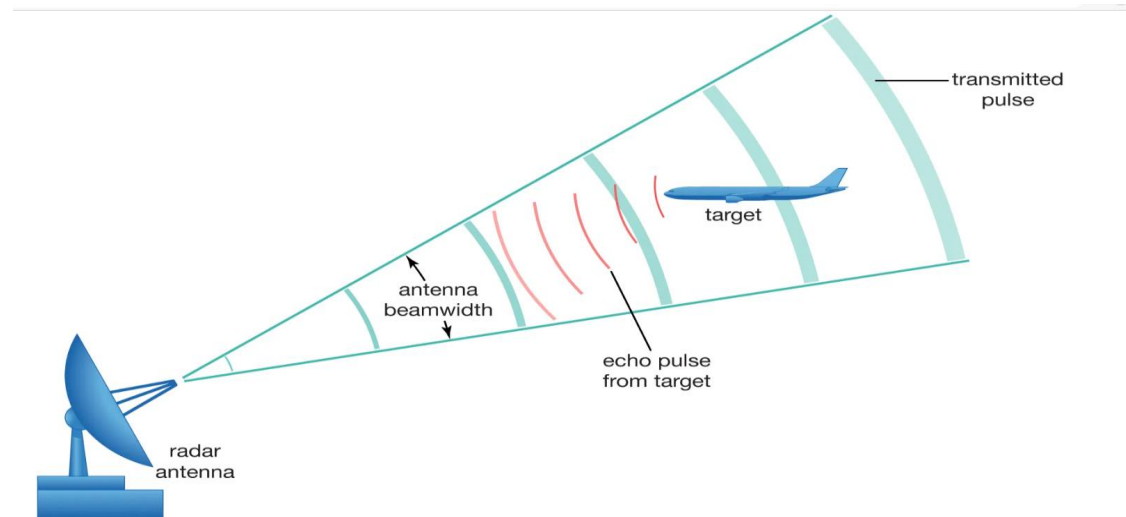


Test of Hypotheses

Motivation:



© Encyclopædia Britannica, Inc.

States of a binary detector

- Usual State – No Signal only Noise – Null Hypotheses – H_0
- Special State – Signal detected – Alternate Hypotheses— H_a

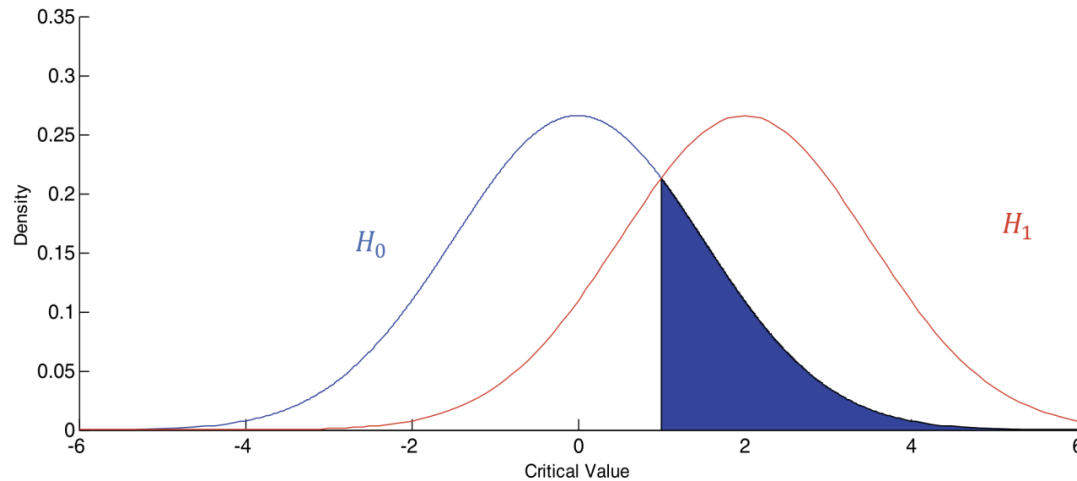
Possible outcomes of a binary detector

1. $D_0|H_0$ —correct decision -- "proper dismissal"
2. $D_1|H_0$ — Type I Error -- "false alarm"
3. $D_0|H_1$ – Type II Error -- "miss"
4. $D_1|H_1$ —correct decision-- "action"

D_0 – decide noise

D_1 – decide signal

Decision Procedure in a detector



Name	Decision	Hypothesis	Probability
Proper dismissal	D_0	H_0	$P_{PD} = P(D_0 H_0) = \int_{\mathcal{Z}_0} p(z H_0)dz$
False alarm	D_1	H_0	$P_{FA} = P(D_1 H_0) = \int_{\mathcal{Z}_1} p(z H_0)dz$
Detection	D_1	H_1	$P_D = P(D_1 H_1) = \int_{\mathcal{Z}_1} p(z H_1)dz$
Miss	D_0	H_1	$P_M = P(D_0 H_1) = \int_{\mathcal{Z}_0} p(z H_1)dz$

$$\mathcal{Z}_0 = \{z : z < \gamma\}, \mathcal{Z}_1 = \{z : z \geq \gamma\}$$

Test of Hypothesis Procedure

1. Formulate the Null hypothesis– H_0
2. Formulate the Alternate hypothesis – H_a
3. Choose an appropriate test statistic(TS) and compute its value from the samples
4. Decide an α – *level of significance or Type I error*
5. Using TS and α compute the rejection region(RR)
6. If the observed value of TS lies in the RR reject H_0 in favor of H_a else state: not able to reject H_0 on the basis of observed evidence
7. Compute the *p-value* from the null distribution of the TS and interpret it

Examples

Usual format:

$$H_0: \mu = \mu_0$$

against one of the alternatives

$H_a: \mu \neq \mu_0$ --- two tailed alternative

$H_a: \mu < \mu_0$ --- lower or left tailed alternative

$H_a: \mu > \mu_0$ --- upper or right tailed alternative

Example 7.1.1

In a coin-tossing experiment, let p be the probability of heads. We start with the claim that the coin is fair, that is, $H_0: p = 1/2$. We test this against one of the following alternatives:

- (a) H_a : The coin is not fair ($p \neq 1/2$). This is a two-tailed alternative.
 - (b) H_a : The coin is biased in favor of heads ($p > 1/2$). This is an upper tailed alternative.
 - (c) H_a : The coin is biased in favor of tails ($p < 1/2$). This is a lower tailed alternative.
-

Few Definitions

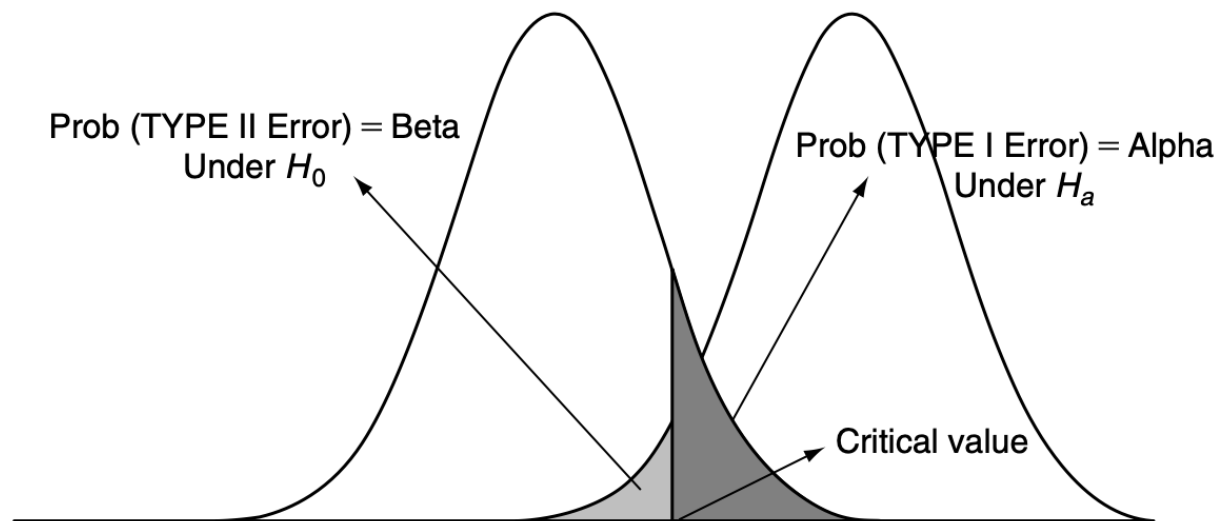
Definition 7.1.2 (a) A **type I error** is made if H_0 is rejected when in fact H_0 is true. The probability of type I error is denoted by α . That is,

$$P(\text{rejecting } H_0 | H_0 \text{ is true}) = \alpha.$$

The probability of type I error, α , is called **the level of significance**.

(b) A **type II error** is made if H_0 is accepted when in fact H_a is true. The probability of a type II error is denoted by β . That is,

$$P(\text{not rejecting } H_0 | H_0 \text{ is false}) = \beta.$$



Examples

Example 7.1.3

A toy store chain claims that at least 80% of girls under 8 years old prefer dolls over other types of toys. After observing the buying pattern of many girls under 8 years old, we feel that this claim is inflated. In an attempt to dispose of this claim, we observe the buying pattern of 20 randomly selected girls under 8 years old, and we observe X , the number of girls who buy stuffed toys or dolls. We wish to test the hypothesis $H_0 : p = 0.8$ against $H_a : p < 0.8$. Suppose we decide to accept the H_0 if $X > 12$ (that is $X \geq 13$). This means that if $\{X \leq 12\}$ (that is $X < 13$) we will reject H_0 .

- (a) Find α .
- (b) Find β for $p = 0.6$.
- (c) Find β for $p = 0.4$.
- (d) Find the rejection region of the form $\{X \leq K\}$ so that (i) $\alpha = 0.01$; (ii) $\alpha = 0.05$.
- (e) For the alternative $H_a : p = 0.6$, find β for the values of α in part (d).

Example 7.1.4

Let X be a binomial random variable. We wish to test the hypothesis $H_0 : p = 0.8$ against $H_a : p = 0.6$. Let $\alpha = 0.03$ be fixed. Find β for $n = 10$ and $n = 20$.